Stobite Compact — How Every Part Works

Oct 3, 2026 · @Ben Sampson

What it is

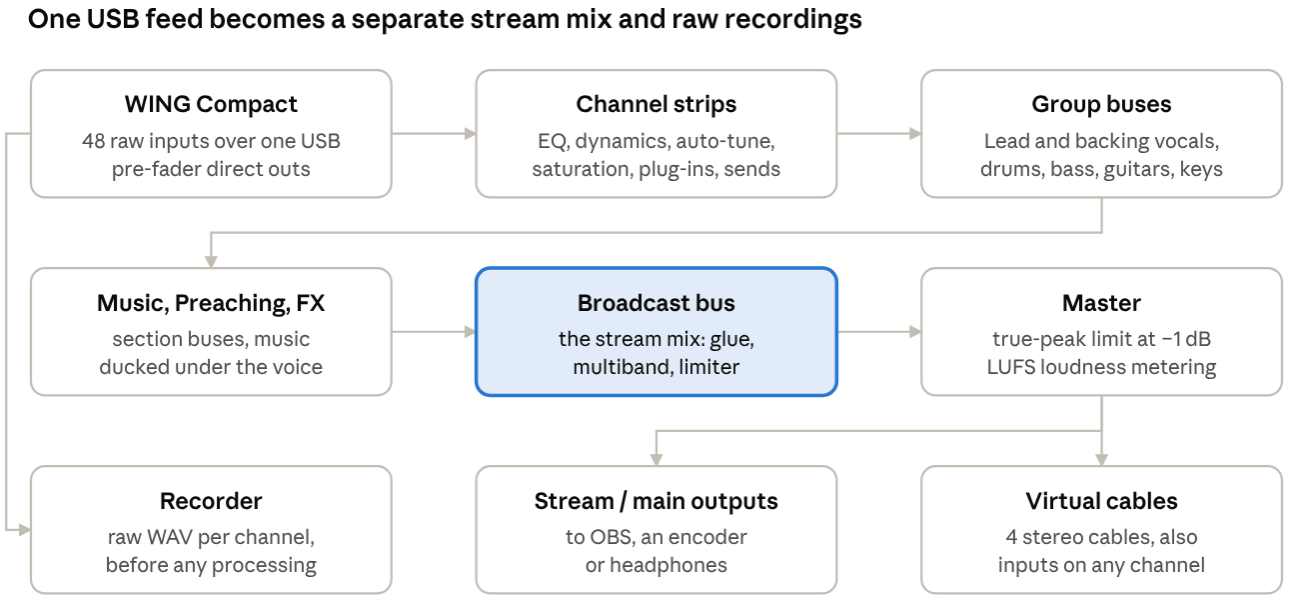
Stobite Compact is a second, independent mixer for the livestream that sits beside the Behringer WING Compact. The WING keeps mixing the room (front of house). The app receives every WING channel raw over one USB cable, records it, and builds a separate broadcast mix for the stream, without ever changing anything on the WING.

It does four jobs in one program:

- **Multitrack recorder and editor**: records up to 48 channels at once, then lets you cut, fade and remix the service afterwards, like Logic or Pro Tools.
- **Broadcast mixer**: channel processing, group buses, reverbs and delays, ducking, and a mastering chain built for worship and preaching.
- **Loudness and quality control**: LUFS, true peak, loudness range, spectrum and mono checks, so the stream lands at the right level on YouTube or Facebook.
- **Remote**: the same screens open on any phone or laptop on the church Wi-Fi, so you can mix from outside the room.

It runs as a desktop app on Mac (DMG) and Windows (installer). It works offline; the internet is only needed to build and download new versions.

Every sound follows one path, from left to right:



signal flow · WING to stream

Recording taps the WING feed on the left, before any processing. Everything else flows through channel strips and buses into the one broadcast mix that becomes the stream.

Hardware and connection

One USB cable carries all the audio: 48 channels into the computer and 48 back out at 48 kHz.

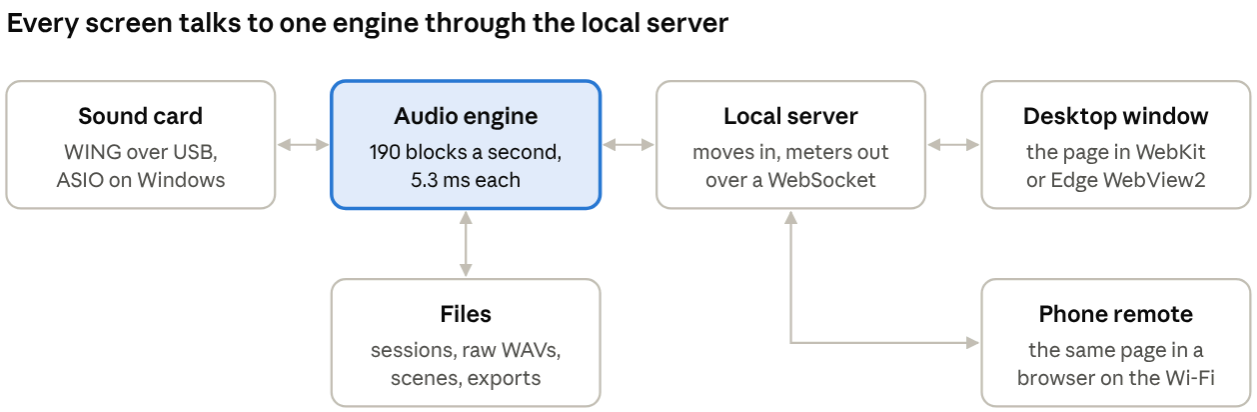
Part	What to use	Why
Cable	USB-A to USB-B, or USB-C to USB-B (USB 2.0 is enough)	USB-B is the square printer plug the WING uses for audio
WING port	The USB-B port on the back panel	The front USB-A socket is only for USB sticks
Length	3 m or less; up to 5 m usually works	Longer runs need an active (powered) extension
Mac driver	None	The WING appears as a 48-channel audio device
Windows driver	Behringer WING USB/ASIO driver	ASIO is the low-latency audio standard on Windows
Ethernet (optional)	WING network port to the church router	Lets the app read the WING's channel names; not needed for audio

On the WING, open ROUTING → Outputs → USB and send the channels' direct outs, set to pre-fader, to USB 1–48. Pre-fader means the WING sends each mic before its own faders and mutes, so moves made for the room never change the stream or the recordings.

Recordings are always raw. The app writes each channel's input to disk before any trim, EQ, compression or fader. You can remix a service later from untouched audio, and nothing done in the app is baked in. The files are 24-bit WAV, one per channel, about 25 GB per hour for all 48 channels; only channels armed for recording are written.

How it is built

The app has three layers: an audio engine that does the sound, a small local server, and an interface that is a web page. The same page runs in the desktop window and in any phone's browser, which is why the remote looks identical.



architecture · how the parts talk

The engine is the only part that touches sound; every screen, local or on a phone, just sends it changes and shows its meters.

- **Audio engine** (Python with NumPy). The sound card calls it about 190 times a second with a block of 256 samples per channel (5.3 ms). In that window it routes inputs, records, runs every processor and sends the result out. Reverbs and third-party plug-ins run on the pedalboard library (built on JUCE, which many commercial plug-ins use).
- **Local server** (aiohttp). It carries every fader move from any screen to the engine, and sends meters back 20 times a second over a WebSocket, a connection that stays open both ways.
- **Interface** (HTML, CSS and JavaScript). On the computer it runs in a native window (WebKit on Mac, Edge WebView2 on Windows). On a phone it is the same page at the address shown in Setup. Every connected screen stays in sync because all of them talk to the one engine.
- **Files.** Everything lives in the *Stobite Compact* folder in your home folder: Sessions (recordings plus a session file), Exports, Scenes and the settings file. Editing is non-destructive: cuts and fades are instructions in the session file, never changes to the WAV recordings. The session saves itself every 10 seconds.

The header bar

The bar across the top is always visible, on every tab.

Control	What it does	Shortcut
Go to start	Moves the playhead to 0:00	Home or Enter
Back / Forward	Jumps 5 seconds	, and .
Play / Pause	Plays the recorded session through the mixer	Space

Control	What it does	Shortcut
Stop	Stops; pressed again, returns to the start	—
Record	Records every armed channel; new takes go after the existing audio	R
Loop	Repeats the selected range (pick it with the Range tool first)	L
Clock	Playhead time in hours, minutes, seconds and milliseconds; turns red while recording	—
Tabs	Edit, Mixer, Routing, Master, Setup (and WING, only if turned on in Setup)	—
Session name	The session you are working in	—
CPU / Idle	How much of each 5.3 ms audio block the engine spends working; Idle means it is resting	—
Two dots	Left: audio engine running (green) or stopped (red). Right: this screen connected to the engine	—
Book icon	Opens the Guide, which explains every control	—

Undo is ⌘Z (Ctrl+Z on Windows). In the Edit tab it undoes edits; in the Mixer, the Master tab or a channel editor it undoes mixer changes. ⌘S saves.

Edit tab: recording and editing

The Edit tab is the multitrack timeline: one row per channel, in the same order as the Mixer, with the recorded audio drawn as waveforms.

Track headers (left column), from left to right: the channel's colour, its number, its instrument icon (click to choose what it is), its name (click to rename), **R** to arm it for recording, **M** mute, **S** solo, **A** automation mode, and a small level meter. Drag a header up or down to reorder; the Mixer follows.

Recording. Arm the channels you want (or **Arm all**) and press record. Each armed channel writes its raw input to its own file while you watch red blocks grow. Stop saves the take as clips on each track.

Tools and edits:

Tool or button	What it does
Pointer (V)	Select clips (Shift adds), drag to move along time or to another track, drag an edge to trim
Blade (B)	Click a clip to cut it in two; Shift-click cuts every track at that point
Range (G)	Drag across time and tracks to select a section
Split (S)	Cuts the selected clips, or all clips, at the playhead
Delete with Ripple	Removes the selected range from every track and closes the gap, e.g. to cut out the offering
Marker (M)	Drops a named marker at the playhead; ← and → jump between markers
Fades	Select a clip and drag the yellow squares at its top corners
Clip bar (bottom)	Clip gain, fade lengths, mute clip, duplicate, delete
Zoom	+, −, 0 to fit, or ⌘/Ctrl and scroll
Used tracks only	Hides empty tracks
Import or drag-and-drop	Drops WAV, AIFF, FLAC or MP3 files onto the track and time where you let go

Automation view. The Automation menu overlays curves on the tracks: volume, pan, mute or one of the four sends. Lanes for every bus, effect return and the master appear underneath. Click to add a point, drag to move it, Alt-click to delete it.

Export gives three choices:

- **Stereo mix**, as WAV or MP3, run through the whole mixer including automation.
- **Stereo mix at a loudness target**, e.g. −14 LUFS for YouTube. The app plays the mix through once to measure it, sets the level, then renders it through the limiter.
- **Separate tracks**, one WAV per channel with edits applied, to open in Logic, Pro Tools or Cubase.

Mixer tab: the channel strip

The Mixer shows one vertical strip per channel, like a console. The buttons across the top switch layers: Inputs (all channels), banks of 16, Buses and FX. The Master strip stays pinned on the right, and saved scenes appear as quick buttons beside the layer buttons.

On each strip, top to bottom: the instrument icon (click it to say what the channel is), channel number, name, input (e.g. IN 5), output bus (e.g. → Drums), a small EQ curve, badges showing which processors are on (click a badge to switch it), a gain-reduction bar, four send knobs, pan, **M** mute, **S** solo, **A** automation, the fader with its meter, and the level in dB. Click the name or EQ curve to open the full channel editor. Drag a strip onto another to reorder.

The channel editor shows the processors in the order the sound passes through them. The breadcrumb at the top traces the whole path, e.g. IN 1 → Lead Vocal → Lead Vocal bus → Broadcast → Master → Stream out. Each **?** opens the Guide at that topic.

Step	What it does	Main controls	Typical use
1. Input	Picks the source: WING USB 1–48, a virtual cable, or none	Input, output bus, instrument, colour	Point any strip at any input
2. Gain / trim and phase	Sets the level entering the strip; phase flips polarity	Trim ±24 dB, Ø	Aim for peaks of −18 to −10 dBFS
3. Low cut / high cut	Removes rumble below, or fizz above, a frequency	Frequency, slope 12, 24 or 48 dB/octave	Low cut ~100 Hz on vocals, keep lows for kick and bass
4. Parametric EQ	Boosts or cuts four frequency ranges	Frequency, gain, Q (width), type: bell, shelves, notch, high-pass, low-pass	Drag points on the graph; the blue haze behind is the live sound
5. Auto-tune	Pulls notes to the song's key (its own section below)	Key, speed, amount, voice	Lead and backing vocals only
6. Gate / expander	Turns the channel down when it falls below a threshold	Threshold, range, attack, release	Drum mics, removing spill
7. Compressor	Turns loud moments down so the level stays steady	Threshold, ratio, attack, release, knee, makeup, mix, max cut	Vocals 3–5:1 with 3–6 dB of reduction; mix below 100% = parallel compression
8. De-esser	Briefly turns the voice down on harsh s and sh sounds	Frequency, threshold, max cut	Every vocal and the preaching mic

Step	What it does	Main controls	Typical use
9. Dynamic EQ	An EQ band that only moves when that range gets loud	Two bands: frequency, Q, threshold, range, static gain, attack, release	Harshness at 3 kHz, mud at 250 Hz, boomy bass notes
10. Transient shaper	Adds or removes the hit and the ring of a sound	Attack ±12 dB, sustain ±12 dB	More snap on kick and snare, tighter toms
11. Saturation	Adds gentle harmonics for warmth and density	Type: tape, tube, console, soft clip; drive; mix	Fullness on small phone speakers
12. Plug-ins	Your own VST3 or Audio Unit plug-ins	Settings as sliders, bypass	Anything the built-in tools don't cover
13. Fader, pan, sends	Sets the level, places it left to right, and sends a share to the effects	Fader, pan, four sends	Sends follow the fader (post-fader)

The editor also has Copy, Paste and Reset for whole strips, a list of starting presets, and ‹ › arrows to step to the next channel.

Buses, effects and the master

A bus is a sub-mix: several channels added together so you can ride and process them as one. Channels feed group buses, group buses feed bigger buses, and everything ends at the Master, which is the stream.

Bus	Fed by	Feeds
Lead Vocal	Worship leader and lead singer channels	Broadcast
Backing Vocals	Backing singers and choir	Broadcast
Drums	Kick, snare, toms, hi-hat, overheads, room	Music
Bass	Bass and synth bass	Music
Guitars	Electric and acoustic guitars	Music
Keys	Piano, keys, pads, organ	Music
Other Instruments	Strings, brass, percussion extras	Music

Bus	Fed by	Feeds
Music	The five instrument buses, plus unlabelled channels	Broadcast
Preaching	Preacher and host mics	Broadcast
FX Returns	The four reverbs and delays	Broadcast
Broadcast	Everything above	Master

You can add, rename, re-route or remove buses. The app stops you routing a bus into itself.

Every bus has the same toolkit, and its processors run in an order you choose: drag them in "Processing order", and each ON switch bypasses one instantly.

- EQ with low and high cuts
- Compressor, with a mix knob for parallel compression and an optional sidechain (it can react to a different bus or channel)
- Saturation
- Dynamic EQ
- Multiband compressor (three bands)
- Width and balance
- Limiter
- Ducking
- Sends to the effects
- Plug-ins

Ducking turns a bus down automatically while another source is active. Examples: the Music bus dips 1–2 dB when the Lead Vocal sings, or 3–6 dB under the preacher. You set the source, threshold, maximum dip, ratio, attack and release.

The four effects work as send and return: channels send a share, the effect runs once, and its output returns through the FX Returns bus. That saves processing and puts everyone in the same space.

Effect	Character	Default use
Vocal Plate	Short, bright, 25 ms pre-delay	A touch on the lead vocal
Vocal Hall	Large, spacious, 40 ms pre-delay	Backing vocals and worship ambience

Effect	Character	Default use
Instrument Verb	A room for the band	Snare, guitars, keys
Delay	Tempo-synced echoes: 1/2, 1/4, dotted 1/4, 1/8, dotted 1/8, triplet, 1/16, or milliseconds; ping-pong and tap tempo	Ducked by the Lead Vocal, so it blooms between lines

The reverbs have pre-delay, decay, size, damping, width, early reflections and a low and high cut on the return. Any effect can be switched to chorus, phaser or doubler.

The Master strip is the last stop before the stream. It has the same chain as a bus (EQ, compressor, saturation, dynamic EQ, multiband, width) and ends with a true-peak limiter set to −1 dB, so the stream cannot clip.

Routing tab and virtual cables

The Routing tab is a live map of where every signal goes. Columns run left to right: channels (with their input), group buses, then Music, Vocals, Preaching and FX, then Broadcast, then Master with its outputs and cables. Coloured lines join each item to its destination.

- Change any "→" dropdown to re-route; the map redraws.
- Each bus card shows how many sources feed it and which processors are on.
- **Load worship broadcast template** puts gentle starting settings on the vocal, drum, music, preaching and broadcast buses and the master in one click. Channel settings are left alone.
- Channels with default names are hidden until you name them or pick an instrument; tick "Show unnamed channels" to see all 48.

Virtual cables are four stereo connections that act as a speaker and a microphone at the same time, like VB-Cable. You send something into a cable (the Master, a bus or a channel), and it comes back out as an input called "Cable 1 L/R" that any channel can use. By default Cable 1 carries the Master.

Inside the app the cables just work. To reach other programs such as OBS, Zoom or YouTube Studio, attach a system loopback device in Setup: BlackHole 16ch on Mac (Cable 1 = channels 1–2, Cable 2 = 3–4, and so on) or VB-Audio Virtual Cable on Windows. In OBS you then add that device as the audio source.

A channel with no recorded audio keeps listening to its live input even during playback, so a cable or a live mic is never silenced by pressing play.

Master tab: loudness and quality control

The Master tab answers one question: will the stream sound right, and at the right level, on every phone and laptop?

YouTube, Facebook and Spotify measure loudness in **LUFS** (loudness units relative to full scale), a measure of loudness as ears hear it, not just peaks. Each platform turns loud streams down to its target, so mixing louder only costs dynamics. The meters follow the ITU BS.1770 and EBU R128 broadcast standards; a test tone that should read −20.0 LUFS reads exactly −20.0.

Reading	What it means	Good range
Momentary	Loudness over the last 0.4 s	Moves constantly; for reacting
Short-term	Loudness over the last 3 s; the one to watch while mixing	Near the target on full choruses
Integrated	Average over the whole service, with silence ignored; what platforms use	Within 1 LU of the target (shown green)
True peak (max)	Highest point between samples, measured at 4× resolution; this is what clips after encoding	At or below −1 dBTP
Loudness range	How much loudness varies across the service	6–12 LU feels alive; under 4 is squashed
RMS	Average power of the signal	For reference

Targets run from −10 LUFS (an energetic livestream) through −14 (YouTube) and −16 (Apple Music and podcasts) to −23 (broadcast TV). The graph under the readings shows short-term loudness over time against a dashed target line.

Other tools on the tab:

- **Spectrum**: how much of each frequency is in the mix, with the master EQ curve on top.
- **Output meters**: peak per side, turning red at clipping.
- **Phase correlation**: +1 means solid mono-safe sound, around 0 very wide, below 0 parts cancel on a mono phone speaker. A red warning appears if it stays negative.
- **Listen check**: Stereo, Mono, Side, Left or Right. To keep these off the stream, set separate monitor outputs in Setup.
- **Mastering chain**: tone EQ → glue compressor → colour (saturation, width, balance, multiband, dynamic EQ) → limiter, with gain-reduction bars.

- **Auto-level to target**: measures what has been playing and sets the output level toward the target.
- **Export mastered mix**: renders the session at the chosen target exactly.

Auto-tune

Auto-tune corrects a singer's pitch live, on the broadcast only; the recording stays untouched. It works in three steps, about 100 times a second:

1. **Hear the note.** It takes the last 40 ms of the voice and finds how often the waveform repeats, using the YIN method, a standard pitch-detection technique. That gives the frequency, e.g. 226 Hz.
2. **Pick the right note.** It turns the frequency into a musical note and finds the nearest note that belongs to the song's key and scale. 226 Hz is 46 cents sharp of A, a cent being 1/100 of a semitone, so in A major the target is A (220 Hz).
3. **Move the voice there.** It glides the pitch toward the target at the retune speed, using a short delay line that is read slightly faster or slower to raise or lower the pitch. It adds about 15 ms of delay, only on tuned channels.

Setting	What it does	Suggestion
Song key and scale	The notes it is allowed to pull to: major, minor, harmonic minor or pentatonic. Shared by all tuned channels	Set it per song; the wrong key pulls singers to wrong notes
Notes	"Song key", or "Any note" for the nearest semitone	Any note when you don't know the key
Voice	Soprano, alto, tenor, baritone or any; limits the range it listens in	Stops band bleed confusing it
Retune speed	How fast a note is pulled in: 0 ms is instant (the robotic effect), 60–120 ms keeps slides and vibrato	Natural 80 ms, Polished 30 ms, Hard 0 ms
Amount	How much of the way it corrects	50–70% keeps the human character

The readout shows the note, how far off it was (green within 10 cents, yellow within 25, red beyond) and how much it is correcting. It is designed for small corrections, up to 2 semitones, and for vocals only.

Instruments, presets, scenes and automation

Instrument library. Click a channel's icon to pick from 27 sources: lead vocal, worship leader, backing vocal, choir, preacher, host, kick, snare, tom, hi-hat, overheads, room mic, percussion, bass, synth bass, electric and acoustic guitar, piano, keys, worship pad, organ, strings, brass, tracks/playback, video, click and audience. One click sets the icon, name and colour, routes the channel to the right bus, and (if ticked) loads a starting preset.

Presets are gentle starting points, never locked. There are 21 for channels, including Lead Vocal, Worship Vocal, Background Vocal, Preaching Mic, Kick, Snare, Toms, Overheads, Bass, Electric and Acoustic Guitar, Piano, Keys and Worship Pad. There are 6 for buses: Vocal, Drum, Music, Speech, Broadcast and Worship Master. Each sets filters, EQ, dynamics, saturation and sends, and every value stays adjustable.

Scenes are snapshots of the whole mixer, such as Pre-Service, Worship, Worship Quiet, Prayer, Preaching, Altar Call and Closing. Saved scenes appear as one-click buttons in the Mixer.

- **Recall changes** decides what a recall touches: faders and pan, mutes, EQ, dynamics, sends, effect settings, routing. For example, recall only faders and mutes when moving from Worship to Preaching, without undoing an EQ fix made earlier.
- **Recall-safe channels** are never changed by scenes, e.g. the preaching mic you are riding by hand.
- **Load all** replaces everything.

Automation records moves over time so the mix performs itself on playback. Every channel, bus, effect and the master has an **A** button with three modes:

Mode	Button	What happens
Off	Grey	Automation is ignored
Read	Green	Plays back recorded moves; faders move by themselves
Touch	Yellow	While playing, any move you make is recorded: volume, pan, mute, sends, bus and effect levels, ducking amount and plug-in settings. Let go and it returns to what was recorded

Use it for intentional transitions: lifting the hall into a bridge, pulling the band down for prayer, opening the delay on a last chorus. Draw or edit the curves in the Edit tab.

Setup tab and the phone remote

Card	What it controls
Audio interface	Input device (Auto finds the WING; a ★ marks it), output device, sample rate (48 kHz is the WING's default) and buffer size (256 recommended; 512 if you hear clicks)
Outputs	Which outputs carry the stream mix, and optional separate monitor (headphone) outputs so listen checks never affect the stream. Live monitoring lets you hear inputs through the mixer when not playing back. USB returns can send processed channels back to the WING, off by default because it adds a few milliseconds of delay
Channels from the WING	**Copy names from WING** reads the console's channel names and colours over the network, read-only. **Match WING order** sets channel 1 = USB 1 and the original order
Virtual cables	Attaches BlackHole or VB-Cable so the four cables reach other programs
WING console control	Off by default. Shows the WING tab, which can move the console's own faders and mutes. The app never changes the WING unless this is on
Phone remote	The Wi-Fi address to open on a phone, and an optional PIN for guest Wi-Fi
Sessions	Create, open and find sessions
Connecting the WING Compact	The cable and routing steps from this document
Storage	Where files are saved and how much space recording takes

The phone remote is the full app in a phone browser, at the address shown in Setup (e.g. http://192.168.1.35:8730). Phone and computer must be on the same Wi-Fi. Every screen controls the same engine and updates the others live, so you can walk outside the room, adjust the stream mix, and walk back. On a phone the mixer scrolls sideways. The computer never needs the PIN.

Efficiency, delay and reliability

The app does work only where there is sound, and rests otherwise.

Situation	What the engine does	Measured cost
Full worship mix playing (15 processed channels, 11 buses, 4 effects, master)	Everything runs	About 27% of each 5.3 ms audio block
Some sections silent, e.g. only the preacher talking	Silent buses, effects and channel filters are skipped	Less than the full mix
Idle: nothing playing, all inputs below −70 dB for 3 s, effect tails finished	The engine sleeps; the header says Idle	Under 1% of the block; about 5% of one CPU core for the whole app
Memory	Only the needed parts of each library are loaded	About 50–80 MB

The engine wakes on the very first block of sound, so nothing is cut off. When the app is idle, meters update 4 times a second instead of 20, and screens redraw only when something changes.

Delay. The room never hears this app, because the WING mixes the room, so delay only matters for the stream. Audio moves in 5.3 ms blocks at the recommended buffer of 256. The limiter's lookahead adds 1.5 ms and auto-tune adds about 15 ms on tuned channels only. Raise the buffer to 512 if you hear clicks.

Reliability:

- Recording writes to disk on its own thread, so a busy screen can't interrupt it; the header warns if the disk ever falls behind.
- The session saves every 10 seconds and when recording stops.
- Edits never change the recordings, and undo keeps the last 200 edits plus mixer changes.
- The output limiter caps the stream at −1 dB true peak.
- If anything in the audio path fails, the app outputs silence instead of noise.

Building and releasing updates

The source code lives in a private GitHub repository, and one Terminal command ships a new version:

```
update compact "What changed"
```

1. **Checks the code** for mistakes, and stops before anything is published if it finds one.
2. **Picks the next version number** (v1.1.0 → v1.1.1; add --minor for v1.2.0), commits every change and pushes it to GitHub.
3. **Tags the version**, which tells GitHub to build.
4. **GitHub builds both apps** on its own Mac and Windows machines in about 3 minutes: the Mac DMG and the Windows installer (Stobite-Compact-Setup.exe).
5. **Downloads both** to Downloads → Stobite Compact vX.Y.Z.
6. **Offers to install** the new Mac app into Applications, then opens the folder.

Old versions stay on the GitHub Releases page, so you can always go back.

Installing on other computers:

- **Mac:** open the DMG and drag the app to Applications. The first time, right-click it and choose Open, because it isn't signed with a paid Apple developer certificate.
- **Windows:** run the Setup.exe; at the SmartScreen warning click More info → Run anyway, for the same reason. The installer adds Start Menu and desktop shortcuts and opens the Windows firewall for the phone remote.
- Install the WING ASIO driver on Windows before first use.

Limits and quick answers

What it does not do yet:

- It hasn't been tested on a real WING. Testing used a virtual audio device and generated tones; the first service is the real test.
- The Windows version builds successfully on GitHub but hasn't been run on a Windows PC yet.
- Plug-ins appear as sliders; their own windows don't open.
- No MIDI, no virtual instruments and no time-stretching.
- Auto-tune is for small corrections (up to 2 semitones) and doesn't preserve formants (the natural character of the voice) on large shifts.
- Sending the cables to OBS or Zoom needs BlackHole (Mac) or VB-Cable (Windows) installed.
- The apps aren't signed with paid developer certificates, so first launch shows a warning.

Questions people will ask:

Question	Answer
Does it change the room sound?	No. The WING mixes the room; this app only makes the stream mix and recordings.
What if the computer crashes during a service?	The room is unaffected. Audio written to disk up to that moment is kept, and the session was saved within the last 10 seconds.
Are the recordings processed?	No. They are raw, pre-fader inputs, so the service can be remixed later.
How many channels?	48 in and 48 out over one USB cable.
Can I mix from my phone?	Yes, from any phone or laptop on the same Wi-Fi, with an optional PIN.
How loud will the stream be?	You choose the target, usually −14 LUFS, and the meters and export hit it; true peak is capped at −1 dB.
Does it need the internet?	Only to build and download updates.
Can it open in Logic or Pro Tools?	Export separate tracks as WAV files and open them there.
Mac or Windows?	Both, from the same code: a DMG for Mac and an installer for Windows.